PG No. 31 T_{d} $\bar{4}3m$ [cubic] (internal polar dipole)

* Harmonics for rank 0

$\vec{\mathbb{Q}}_{0}^{(1,1)}[q](A_1)$

** symmetry

1

** expression

$$\frac{\sqrt{3}Q_x x}{3}+\frac{\sqrt{3}Q_y y}{3}+\frac{\sqrt{3}Q_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_{1,1}^{(1,0)}[q](T_1)$, $\vec{\mathbb{G}}_{1,2}^{(1,0)}[q](T_1)$, $\vec{\mathbb{G}}_{1,3}^{(1,0)}[q](T_1)$

** symmetry

x

y

z

** expression

$$\frac{\sqrt{2}Q_y z}{2}-\frac{\sqrt{2}Q_z y}{2}$$

$$-\frac{\sqrt{2}Q_x z}{2}+\frac{\sqrt{2}Q_z x}{2}$$

$$\frac{\sqrt{2}Q_x y}{2}-\frac{\sqrt{2}Q_y x}{2}$$

$\vec{\mathbb{Q}}_{1,1}^{(1,-1)}[q](T_2)$, $\vec{\mathbb{Q}}_{1,2}^{(1,-1)}[q](T_2)$, $\vec{\mathbb{Q}}_{1,3}^{(1,-1)}[q](T_2)$

** symmetry

x

y

z

** expression

Q_x

Q_y

Q_z

$\vec{\mathbb{Q}}_{1,1}^{(1,1)}[q](T_2)$, $\vec{\mathbb{Q}}_{1,2}^{(1,1)}[q](T_2)$, $\vec{\mathbb{Q}}_{1,3}^{(1,1)}[q](T_2)$

** symmetry

x

y

z

** expression

$$\frac{\sqrt{10}Q_x\left(2x^2-y^2-z^2\right)}{10}+\frac{3\sqrt{10}Q_y xy}{10}+\frac{3\sqrt{10}Q_z xz}{10}$$

$$\frac{3\sqrt{10}Q_x xy}{10}-\frac{\sqrt{10}Q_y\left(x^2-2y^2+z^2\right)}{10}+\frac{3\sqrt{10}Q_z yz}{10}$$

$$\frac{3\sqrt{10}Q_x xz}{10}+\frac{3\sqrt{10}Q_y yz}{10}-\frac{\sqrt{10}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_{2,1}^{(1,-1)}[q](E)$, $\vec{\mathbb{Q}}_{2,2}^{(1,-1)}[q](E)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{\sqrt{6}Q_x x}{6} - \frac{\sqrt{6}Q_y y}{6} + \frac{\sqrt{6}Q_z z}{3}$$

$$\frac{\sqrt{2}Q_x x}{2} - \frac{\sqrt{2}Q_y y}{2}$$

$\vec{\mathbb{Q}}^{(1,1)}_{2,1}[q](E)$, $\vec{\mathbb{Q}}^{(1,1)}_{2,2}[q](E)$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{\sqrt{21}Q_x x\left(x^2+y^2-4z^2\right)}{14} - \frac{\sqrt{21}Q_y y\left(x^2+y^2-4z^2\right)}{14} - \frac{\sqrt{21}Q_z z\left(3x^2+3y^2-2z^2\right)}{14}$$

$$\frac{\sqrt{7}Q_x x\left(3x^2-7y^2-2z^2\right)}{14} + \frac{\sqrt{7}Q_y y\left(7x^2-3y^2+2z^2\right)}{14} + \frac{5\sqrt{7}Q_z z\left(x-y\right)\left(x+y\right)}{14}$$

$\vec{\mathbb{G}}^{(1,0)}_{2,1}[q](E)$, $\vec{\mathbb{G}}^{(1,0)}_{2,2}[q](E)$
** symmetry

$$-\frac{\sqrt{3}x^2}{2} + \frac{\sqrt{3}y^2}{2}$$

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$-\frac{\sqrt{2}Q_x yz}{2} - \frac{\sqrt{2}Q_y xz}{2} + \sqrt{2}Q_z xy$$

$$\frac{\sqrt{6}Q_x yz}{2} - \frac{\sqrt{6}Q_y xz}{2}$$

$\vec{\mathbb{G}}^{(1,0)}_{2,1}[q](T_1)$, $\vec{\mathbb{G}}^{(1,0)}_{2,2}[q](T_1)$, $\vec{\mathbb{G}}^{(1,0)}_{2,3}[q](T_1)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}Q_x\left(y-z\right)\left(y+z\right)}{2} - \frac{\sqrt{2}Q_y xy}{2} + \frac{\sqrt{2}Q_z xz}{2}$$

$$\frac{\sqrt{2}Q_x xy}{2} - \frac{\sqrt{2}Q_y\left(x-z\right)\left(x+z\right)}{2} - \frac{\sqrt{2}Q_z yz}{2}$$

$$-\frac{\sqrt{2}Q_x xz}{2} + \frac{\sqrt{2}Q_y yz}{2} + \frac{\sqrt{2}Q_z\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{Q}}^{(1,-1)}_{2,1}[q](T_2)$, $\vec{\mathbb{Q}}^{(1,-1)}_{2,2}[q](T_2)$, $\vec{\mathbb{Q}}^{(1,-1)}_{2,3}[q](T_2)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}Q_{y}z}{2}+\frac{\sqrt{2}Q_{z}y}{2}$$

$$\frac{\sqrt{2}Q_{x}z}{2}+\frac{\sqrt{2}Q_{z}x}{2}$$

$$\frac{\sqrt{2}Q_{x}y}{2}+\frac{\sqrt{2}Q_{y}x}{2}$$

$\vec{\mathbb{Q}}_{2,1}^{(1,1)}[q](T_2)$, $\vec{\mathbb{Q}}_{2,2}^{(1,1)}[q](T_2)$, $\vec{\mathbb{Q}}_{2,3}^{(1,1)}[q](T_2)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{5\sqrt{7}Q_{x}xyz}{7}-\frac{\sqrt{7}Q_{y}z\left(x^{2}-4y^{2}+z^{2}\right)}{7}-\frac{\sqrt{7}Q_{z}y\left(x^{2}+y^{2}-4z^{2}\right)}{7}$$

$$\frac{\sqrt{7}Q_{x}z\left(4x^{2}-y^{2}-z^{2}\right)}{7}+\frac{5\sqrt{7}Q_{y}xyz}{7}-\frac{\sqrt{7}Q_{z}x\left(x^{2}+y^{2}-4z^{2}\right)}{7}$$

$$\frac{\sqrt{7}Q_{x}y\left(4x^{2}-y^{2}-z^{2}\right)}{7}-\frac{\sqrt{7}Q_{y}x\left(x^{2}-4y^{2}+z^{2}\right)}{7}+\frac{5\sqrt{7}Q_{z}xyz}{7}$$

* Harmonics for rank 3
$\vec{\mathbb{Q}}_{3}^{(1,-1)}[q](A_1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$Q_{x}yz+Q_{y}xz+Q_{z}xy$$

$\vec{\mathbb{Q}}_{3}^{(1,1)}[q](A_1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{15}Q_{x}yz\left(6x^{2}-y^{2}-z^{2}\right)}{6}-\frac{\sqrt{15}Q_{y}xz\left(x^{2}-6y^{2}+z^{2}\right)}{6}-\frac{\sqrt{15}Q_{z}xy\left(x^{2}+y^{2}-6z^{2}\right)}{6}$$

$\vec{\mathbb{G}}_{3}^{(1,0)}[q](A_2)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{5}Q_{x}x\left(y-z\right)\left(y+z\right)}{2}-\frac{\sqrt{5}Q_{y}y\left(x-z\right)\left(x+z\right)}{2}+\frac{\sqrt{5}Q_{z}z\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,-1)}[q](T_1)$, $\vec{\mathbb{Q}}_{3,2}^{(1,-1)}[q](T_1)$, $\vec{\mathbb{Q}}_{3,3}^{(1,-1)}[q](T_1)$
** symmetry

$$\frac{\sqrt{15}xy^{2}}{2}-\frac{\sqrt{15}xz^{2}}{2}$$

$$-\frac{\sqrt{15}x^{2}y}{2}+\frac{\sqrt{15}yz^{2}}{2}$$

$$\frac{\sqrt{15}x^{2}z}{2}-\frac{\sqrt{15}y^{2}z}{2}$$

** expression

$$\frac{Q_x\,(y-z)\,(y+z)}{2}+Q_y xy-Q_z xz$$

$$-Q_x xy-\frac{Q_y\,(x-z)\,(x+z)}{2}+Q_z yz$$

$$Q_x xz-Q_y yz+\frac{Q_z\,(x-y)\,(x+y)}{2}$$

$\vec{\mathbb{Q}}^{(1,1)}_{3,1}[q](T_1)$, $\vec{\mathbb{Q}}^{(1,1)}_{3,2}[q](T_1)$, $\vec{\mathbb{Q}}^{(1,1)}_{3,3}[q](T_1)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{15}Q_x\,(y-z)\,(y+z)\left(6x^2-y^2-z^2\right)}{12}-\frac{\sqrt{15}Q_y xy\left(2x^2-5y^2+9z^2\right)}{12}+\frac{\sqrt{15}Q_z xz\left(2x^2+9y^2-5z^2\right)}{12}$$

$$-\frac{\sqrt{15}Q_x xy\left(5x^2-2y^2-9z^2\right)}{12}+\frac{\sqrt{15}Q_y\,(x-z)\,(x+z)\left(x^2-6y^2+z^2\right)}{12}-\frac{\sqrt{15}Q_z yz\left(9x^2+2y^2-5z^2\right)}{12}$$

$$\frac{\sqrt{15}Q_x xz\left(5x^2-9y^2-2z^2\right)}{12}+\frac{\sqrt{15}Q_y yz\left(9x^2-5y^2+2z^2\right)}{12}-\frac{\sqrt{15}Q_z\,(x-y)\,(x+y)\left(x^2+y^2-6z^2\right)}{12}$$

$\vec{\mathbb{G}}^{(1,0)}_{3,1}[q](T_1)$, $\vec{\mathbb{G}}^{(1,0)}_{3,2}[q](T_1)$, $\vec{\mathbb{G}}^{(1,0)}_{3,3}[q](T_1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{3}Q_y z\left(4x^2-y^2-z^2\right)}{4}-\frac{\sqrt{3}Q_z y\left(4x^2-y^2-z^2\right)}{4}$$

$$\frac{\sqrt{3}Q_x z\left(x^2-4y^2+z^2\right)}{4}-\frac{\sqrt{3}Q_z x\left(x^2-4y^2+z^2\right)}{4}$$

$$-\frac{\sqrt{3}Q_x y\left(x^2+y^2-4z^2\right)}{4}+\frac{\sqrt{3}Q_y x\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{Q}}^{(1,-1)}_{3,1}[q](T_2)$, $\vec{\mathbb{Q}}^{(1,-1)}_{3,2}[q](T_2)$, $\vec{\mathbb{Q}}^{(1,-1)}_{3,3}[q](T_2)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{15}Q_x\left(2x^2-y^2-z^2\right)}{10}-\frac{\sqrt{15}Q_y xy}{5}-\frac{\sqrt{15}Q_z xz}{5}$$

$$-\frac{\sqrt{15}Q_x xy}{5} - \frac{\sqrt{15}Q_y\left(x^2-2y^2+z^2\right)}{10} - \frac{\sqrt{15}Q_z yz}{5}$$

$$-\frac{\sqrt{15}Q_x xz}{5} - \frac{\sqrt{15}Q_y yz}{5} - \frac{\sqrt{15}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}^{(1,1)}_{3,1}[q](T_2)$, $\vec{\mathbb{Q}}^{(1,1)}_{3,2}[q](T_2)$, $\vec{\mathbb{Q}}^{(1,1)}_{3,3}[q](T_2)$

** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{Q_x\left(8x^4-24x^2y^2-24x^2z^2+3y^4+6y^2z^2+3z^4\right)}{12} + \frac{5Q_y xy\left(4x^2-3y^2-3z^2\right)}{12} + \frac{5Q_z xz\left(4x^2-3y^2-3z^2\right)}{12}$$

$$-\frac{5Q_x xy\left(3x^2-4y^2+3z^2\right)}{12} + \frac{Q_y\left(3x^4-24x^2y^2+6x^2z^2+8y^4-24y^2z^2+3z^4\right)}{12} - \frac{5Q_z yz\left(3x^2-4y^2+3z^2\right)}{12}$$

$$-\frac{5Q_x xz\left(3x^2+3y^2-4z^2\right)}{12} - \frac{5Q_y yz\left(3x^2+3y^2-4z^2\right)}{12} + \frac{Q_z\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{12}$$

$\vec{\mathbb{G}}^{(1,0)}_{3,1}[q](T_2)$, $\vec{\mathbb{G}}^{(1,0)}_{3,2}[q](T_2)$, $\vec{\mathbb{G}}^{(1,0)}_{3,3}[q](T_2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\sqrt{5}Q_x xyz + \frac{\sqrt{5}Q_y z\left(2x^2+y^2-z^2\right)}{4} + \frac{\sqrt{5}Q_z y\left(2x^2-y^2+z^2\right)}{4}$$

$$\frac{\sqrt{5}Q_x z\left(x^2+2y^2-z^2\right)}{4} - \sqrt{5}Q_y xyz - \frac{\sqrt{5}Q_z x\left(x^2-2y^2-z^2\right)}{4}$$

$$\frac{\sqrt{5}Q_x y\left(x^2-y^2+2z^2\right)}{4} - \frac{\sqrt{5}Q_y x\left(x^2-y^2-2z^2\right)}{4} - \sqrt{5}Q_z xyz$$